

Assignment No:- 05

Title:- Write a program for analysis of quick sort by using deterministic and randomized variant.

Program Code:-

```
import random
import time

# Deterministic QuickSort
def partition(arr, low, high):
    pivot = arr[high]
    i = low - 1
    for j in range(low, high):
        if arr[j] <= pivot:
            i += 1
            arr[i], arr[j] = arr[j], arr[i]
    arr[i+1], arr[high] = arr[high], arr[i+1]
    return i + 1

def quicksort_deterministic(arr, low, high):
    if low < high:
        pi = partition(arr, low, high)
        quicksort_deterministic(arr, low, pi - 1)
        quicksort_deterministic(arr, pi + 1, high)

# Randomized QuickSort
def randomized_partition(arr, low, high):
    pivot_index = random.randint(low, high)
    arr[pivot_index], arr[high] = arr[high], arr[pivot_index]
    return partition(arr, low, high)

def quicksort_randomized(arr, low, high):
    if low < high:
        pi = randomized_partition(arr, low, high)
        quicksort_randomized(arr, low, pi - 1)
        quicksort_randomized(arr, pi + 1, high)

# Main
if __name__ == "__main__":
    arr = [10, 7, 8, 9, 1, 5]
    print("Original Array:")
    print(" ".join(map(str, arr)))

    # Deterministic QuickSort
    arr1 = arr.copy()
    start = time.perf_counter_ns()
    quicksort_deterministic(arr1, 0, len(arr1)-1)
    end = time.perf_counter_ns()
    print("Sorted Array using Deterministic QuickSort:")
    print(" ".join(map(str, arr1)))
    print(f"Time taken by Deterministic QuickSort: {end - start} ns")

    # Randomized QuickSort
    arr2 = arr.copy()
    start = time.perf_counter_ns()
    quicksort_randomized(arr2, 0, len(arr2)-1)
    end = time.perf_counter_ns()
```

```
print("Sorted Array using Randomized QuickSort:")  
print(" ".join(map(str, arr2)))  
print(f"Time taken by Randomized QuickSort: {end - start} ns")
```

OUTPUT:-

Original Array:

10 7 8 9 1 5

Sorted Array using Deterministic QuickSort:

1 5 7 8 9 10

Time taken by Deterministic QuickSort: 22800 ns

Sorted Array using Randomized QuickSort:

1 5 7 8 9 10

Time taken by Randomized QuickSort: 29400 ns